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in partial fulfilment of the requirements for the Degree of

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BLOOD CANCER PREDICTION IN ACUTE LYMPHOBLASTIC LEUKAEMIA

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Abstract

The image processing for the evolution of data sets helped to integrate the prediction process of blood cancer in terms of Acute Lymphoblastic Leukaemia (ALL) in medical application. This research has aimed to develop an ML model for the image processing in prediction of blood cancer. The use of machine learning algorithms by the Python programming language has helped to evaluate the concepts regarding significant implications of the blood cancer prediction process with better accuracy. The use of Deep learning models, such as Resnet Inception V3, MobileNet and CNN with the libraries of python language has helped to illustrate significant ways for prediction of blood cancer in medical application. The deep learning algorithms l has been used to classify cancer cells and healthy blood cells for identifying every stage of blood cancer concerning enhancements of accuracy and adequacy level in cancer detection process. NumPy, Pandas and Matplotlib of Python have been used for the code generation and development of Deep learning models for integration of image processing that led to the prediction of blood cancer with accuracy and efficiency. Among all the deep learning models, the MobileNet V2 model has been the best machine learning model due to the higher accuracy and highest efficiency in prediction of ALL. However, limited used of qualitative data and human centric data has impacted reliability, validity and clarity of research findings. This research has suggested the use of a mixed method concerning the use of qualitative and quantitative data with perception of healthcare professionals and real time instances for the research integration.

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1. Introduction

1.1 Background of the study

Blood cancer has been a significant threat in the healthcare scenario, over the past few years. In this scenario, leukaemia is the most severe malignancy in the blood, which is diagnosed, due to the fast proliferation of aberrant blood cells (Garg et al., 2021). There are also other two types of blood cancer, such as Lymphoma and Myeloma **(*Refer to Figure 1*)**. In the current scenario, the blood cancer diagnosis process is expensive and time-consuming, which includes different tests and consultations with doctors. Acute Lymphoplastic Leukaemia is a type of blood cancer that originates from bone marrow and effects the immature lymphoid cells. It progresses rapidly and leads to the overproduction of abnormal white blood cells. As stated by Yi et al. (2020), Acute lymphoblastic leukaemia (ALL) includes the development of excessive immature lymphocytes affecting the health of individuals. A record total of 6,550 new cases and 1,330 deaths have been enlisted in 2024, which have been caused by suffering from ALL (American Cancer Society, 2023). The Automated diagnosis system has been important in the current scenario for faster diagnosis and accurate prediction of blood cancer so that the possibility of cancer can be prevented in the earlier stage (Varlamova et al., 2024). The computer-aided diagnosis system generally speeds up the ALL cancer diagnosis process significantly and, in these analyses, the medical images are used as the input in the machine learning models.

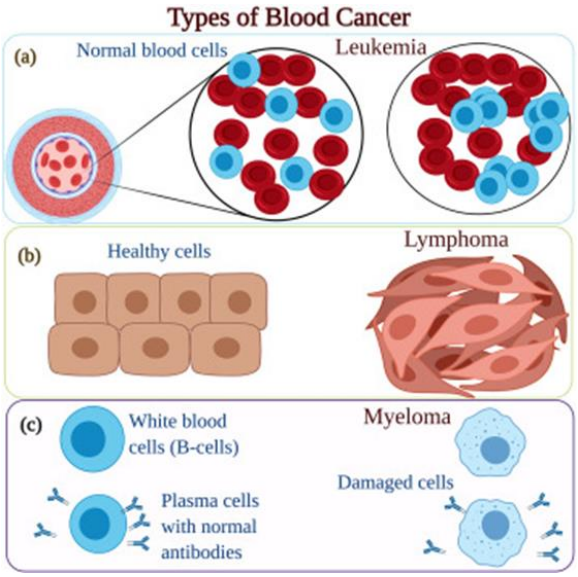

Figure 1: Different types of Blood Cancer

(Source: Garg et al., 2021)

Multiple negative side effects and reduced life quality can occur due to failure in blood cancer prediction. Along with that, there is a lack of accurate prediction for blood cancer and that leads to mistreatment and neglected cases (Rupapara et al., 2022). In this scenario, the prediction models backed by machine learning and artificial intelligence are responsible for the characteristic classification depending on the age. There are different technologies to predict blood cancer including medical images and machine learning. Further, cancer detection using machine learning detects the possibilities of acute lymphoblastic leukaemia in an early phase (Zhang, Shi and Wang, 2023). In the Image classification, the image dataset needs to be prepared as the annotated dataset by defining the labels and the path of the images in the local system. Further, the completed annotated dataset is important for repeated research investigations and data analysis. In the ALL cancer detection by machine learning, the performance of the machine learning model is enhanced by the proper data pre-processing and the data augmentation. Image classification through machine learning comes under big data analysis, as the dataset size is big and it takes a larger time for analysis.

1.2 Problem Statement

Lack of proper cancer recognition and early-stage cancer prediction processes in healthcare practices impacts the proportion of patient diagnoses. The mortality rate of the patients and proper treatment initiation are highly impacted by the lack of an appropriate blood cancer prediction system and the use of conventional methods of predicting cancer (Hunter, Hindocha and Lee, 2022; Batis et al., 2021). It reflects an early and accurate prediction of acute lymphoblastic leukaemia, which is a critical aspect while incorporating traditional processes in the medical field. However, despite advanced medical processes in the diagnosis and prediction process, it encompasses significant challenges including the implementation and operation of the prediction models (Thakur et al., 2024). In this aspect, multiple technological factors can limit the determination of cancerous cells in the blood that might fail blood cancer prediction. Therefore, it can be stated that limitations of proper alignment of predictive models, utilisation of advanced technologies, and misdiagnosis are the present challenging context of blood cancer prediction.

1.3 Research Rationale

Conventional cancer prediction with deep learning algorithms by utilising image data often lacks alignment of biological factors that precisely describe the cancer features. Cancer detection by utilising significant blood cells by empowering transfer learning and different image-processing

techniques has effectively explored where genetic and clinical factors are overlooked (Nasir et al., 2023). In this aspect, utilisation of multimodal data including genetic and clinical factors in the image dataset containing acute lymphoblastic leukaemia has allowed the determination of the influential factors of cancer thereby increasing the accuracy of the ML models. The comparison of different ML models and image processing techniques will ensure the precise model selection of blood cancer prediction with strategic incorporation of genetic and clinical information. Therefore, the study is relevant in providing adequate insights into the area of forecasting blood cancer using an image dataset of acute lymphoblastic leukaemia.

1.4 Research Aim, objectives, and questions

1.4.1 Aim

The research aims to predict blood cancer in terms of acute lymphoblastic leukaemia using an image dataset.

1.4.2 Objectives

- To implement an appropriate machine learning model for precise prediction of blood cancer in terms of acute lymphoblastic leukaemia.
- To assess the effectiveness of multiple image processing techniques for enhancing blood cancer in terms of acute lymphoblastic leukaemia prediction accuracy.
- To explore the influence of integrating combined clinical and genetic data in image datasets of acute lymphoblastic leukaemia to enhance the accuracy of blood cancer prediction.
- To analyse the effectiveness of different deep learning algorithms by comparing their performances that involve blood cancer prediction for acute lymphoblastic leukaemia.
- To successfully built a graphical user interface for predicting blood cancer in terms of acute lymphoblastic leukaemia.

1.4.3 Questions

- How can a machine learning model be implemented to predict blood cancer in terms of acute lymphoblastic leukaemia precisely?
- How do different image processing techniques contribute to enhancing blood cancer prediction for acute lymphoblastic leukaemia accuracy?

- What is the influence of multimodal data integration by combining clinical and genetic factors in image datasets containing the image dataset of acute lymphoblastic leukaemia for enhancing blood cancer prediction accuracy?
- Which deep learning models are the most effective blood cancer prediction processes for acute lymphoblastic leukaemia?
- How can a graphical user interface be built successfully to predict blood cancer in terms of acute lymphoblastic leukaemia?

1.5 Significance of the Study

The research is significant in delivering adequate insights into effective machine-learning models and their significance in predicting blood cancer. Moreover, a different perspective on incorporating the clinical and genetic features within the image dataset of blood samples of acute lymphoblastic leukaemia has been analysed and evaluated rather than the existing research papers (Tufail et al., 2021). In this context, the study is significant in providing a contextual understanding of the accuracy rate of cancer prediction with an adequate and comprehensive understanding of the disease. The multimodal approach will improve the model's performance and accuracy as it will capture both the molecular and visible features thereby ensuring enhanced and personalised diagnosis and treatment plans.

1.6 Structure of the Dissertation

Topic	***Description***
Introduction	The chapter includes a brief overview of the research purpose, rationale, significance, and background with proper identification of the research aim and objectives.
Literature review	The chapter includes detailed information extracted based on the research variables with adequate addressing the research gap of existing literature and theoretical alignment.
Methodology	The chapter delivers an overview of the integrated approach to the research including the utilised software, programming language, and entire research design.

Data analysis	The chapter includes the results of the analysed data in Jupyter Notebook including infographics and its analysis.
Finding and discussion	The chapter delivers adequate insights into the analysed data and results with a proper alignment of findings and research objectives.
Conclusion and recommendations	The chapter includes limitations and key optimisation of the research along with recommendations for future research execution with key improvement areas.

Table 1: Structure of the dissertation

2. Literature Review

2.1 Introduction

In this chapter, the review of literature has provided a comprehensive examination of existing research on ALL, particularly focusing on the utilisation of image datasets. This research improved the capability of predicting blood cancer by formulating a machine-learning model that has estimated the discovery effectively. This review also outlined the conceptual framework by evaluating and contrasting various scholarly sources that guided the research. The study has discussed the integration of advanced image processing techniques and ML algorithms that help to enhance the prediction of blood cancer.

2.2 Overview of Blood Cancer Prediction

The prediction of blood cancer has become increasingly crucial in medical research, particularly for early detection and treatment. As mentioned by Rupapara et al. (2022), the machine learning approach of extracting leukaemia microarray gene data has supported the accuracy of diagnosis ***[Refer to Appendix 1]***. Moreover, as stated by Althaf et al. (2023), diagnosis of blood cancer with the aid of the sensor data by the use of artificial neural networks (ANN) yields faster and more cost-effective results. The automated blood cancer detection processes specifically for leukaemia utilised ANNs through a series of essential steps such as designing, placing, collecting, analysing, constructing, training and evaluating data ***(Refer to Figure 2)***. This method has enhanced diagnostic accuracy, aiding medical professionals in timely intervention and treatment. This signifies the importance of integrating diverse methodologies in improving blood cancer reduction.

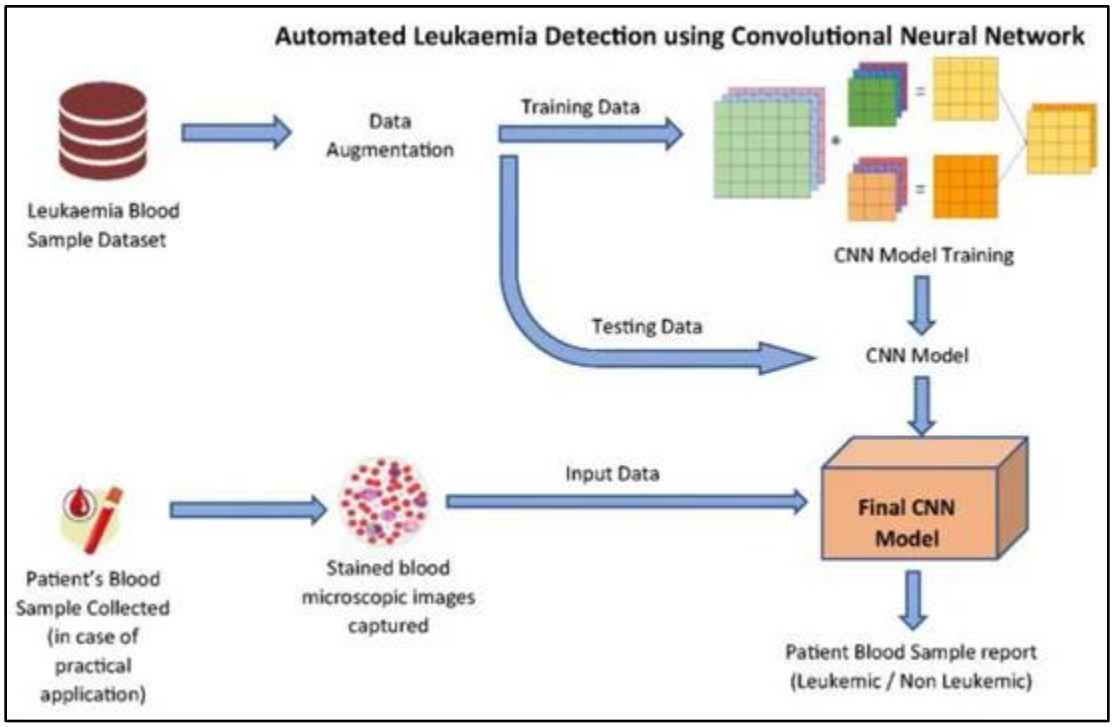

Figure 2:Role of ANN-based artificial intelligence system

(Source: Althaf et al., 2023)

The utilisation of image datasets beyond traditional diagnostic techniques that are increasingly relevant in contemporary healthcare settings. According to Muhsen et al. (2020), advanced techniques such as picture and speech recognition, ML and deep learning helped a lot in diagnosing blood malignancies including myeloma, lymphoma and leukaemia. On the other hand, as described by Saeed et al. (2022), the EfficientNetV2S model is highly effective in predicting acute lymphoblastic leukaemia by analysing image datasets appropriately. These technologies helped healthcare workers to work out the existence of these concerns more proficiently, hence enhancing patient success.

2.3 Concept of machine learning algorithms for cancer detection

Machine learning algorithms play a crucial role in cancer detection, utilising various types of medical data such as images, blood tests and genetic information. Zhang and Wang (2023), highlighted that machine learning algorithms are claimed to find correlations between variations of different blood cancer types explicitly from analytical models including supervised learning, unsupervised learning, deep learning, transfer learning and reinforcement learning. Moreover, as per the view of Taye (2023), feature engineering and scaling are fundamental for the improvement of model performance, since they involve new feature construction and important feature selection

within the data set. In this aspect, the deep learning algorithms were optimised for images from X-rays, MRIs and blood samples while the supervised and unsupervised models usually deal with clinical data. In addition, Alzahrani et al. (2023), argued that exploratory data analysis (EDA) played a crucial role here because it has helped identify trends and patterns that define feature engineering. It has been found that the integration of these diverse MLs has offered a comprehensive framework for improving cancer detection accuracy.

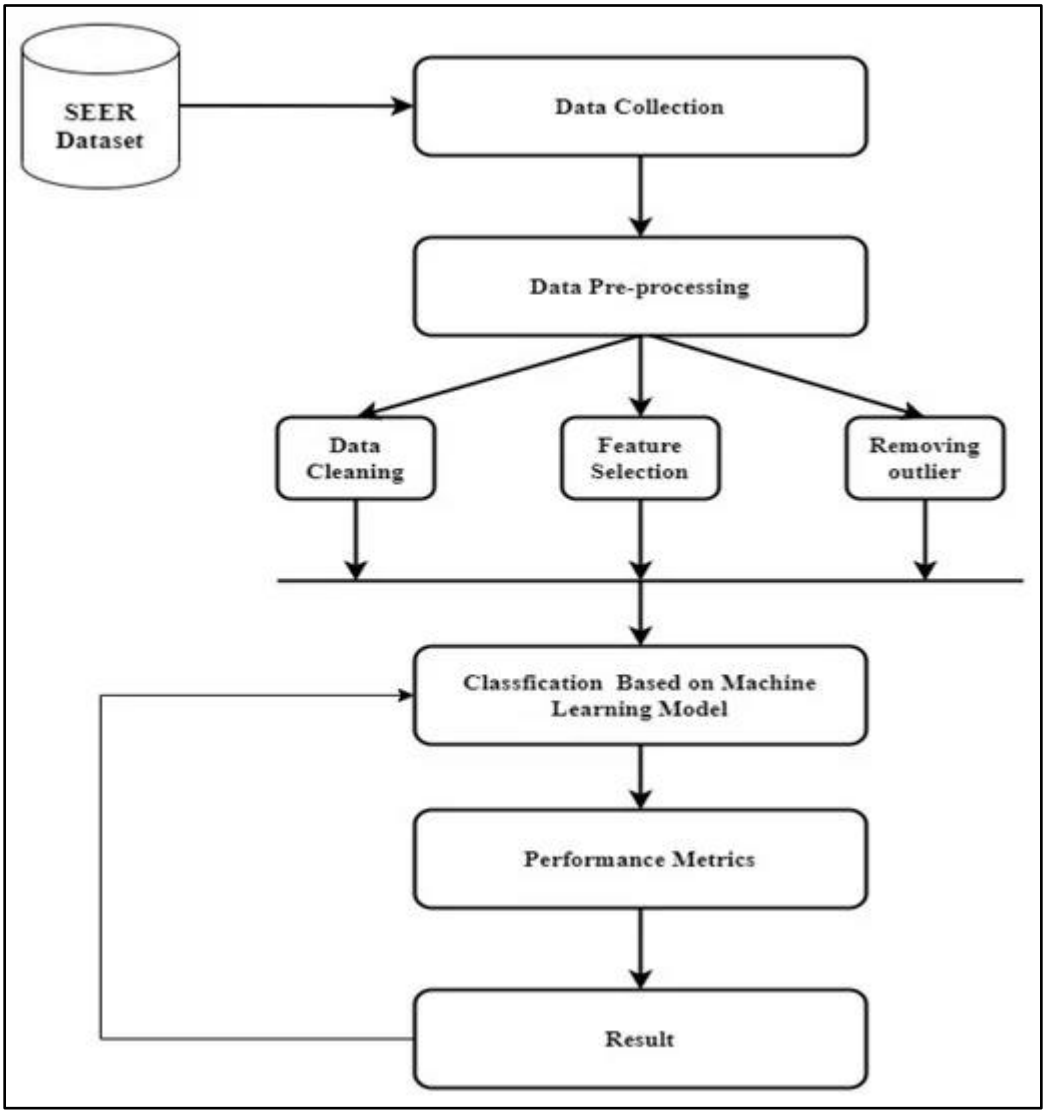

Figure 3:Flowchart of machine learning techniques in cancer research

(Source: Ebrahim, Sedky and Mesbah, 2023)

The flow chart illustrates the ML process of cancer detection including the steps from dataset steering to the final results ***(Refer to Figure 3)***. According to Ebrahim, Sedky and Mesbah (2023), techniques such as deep neural networks, support vector machines and multiple kernel learning

leveraged various data modalities including imaging and electronic health records. For example, as cited by Saeed et al. (2022), acute lymphoblastic leukaemia can be properly determined with the integration of deep learning models to boost early prediction among individuals. Apart from these, these algorithms have automatically selected relevant features that improve accuracy and efficiency in identifying cancerous patterns and facilitating early diagnosis.

2.4 Benefits of python libraries and frameworks for effective blood cancer detection

The involved blood cancers have been easily identifiable with proper ML models and AI analytical tools. It has been most effective to build ML models with the help of Python frameworks and libraries. Taye (2023), suggested that Python libraries and frameworks are dynamic and fast in data preprocessing and EDA along with model building and assessment. As per the view of Siebert, Groß and Schroth (2021), Numpy and pandas are usually used in tabular data only where the data is recorded merely in text form. For example, CancerSEEK analysed large datasets of DNA and protein biomarkers (Douglas, 2020). In the DETECT-A study, samples of blood from about 10,000 women were examined to check the presence of cancer biomarkers (Douglas, 2020). This means that Python libraries and frameworks are crucial for effective data pre-processing and model building in blood cancer detection.

Data pre-processing libraries assist with mathematical operations, dataset manipulation, data cleaning and data segmentation. As stated by Dilhara, Ketkar and Dig (2021), the rich ecosystem of libraries such as Scikit-learn, TensorFlow, Keras and many more, mainly aims to produce strong and reliable models of ML and deep learning. Accordingly, Mazher (2020), opined that the visualisation libraries such as Matplotlib and Seaborn in identifying patterns and trends within a given dataset have been conveniently used through graphical displays. However, Asaad et al. (2023), mentioned that image processing is a crucial part and libraries such as Scikit-image and Python image library (PIL) make the process of improving the datasets easier. In this regard, the EfficientNetV2S model is highly effective in predicting acute lymphoblastic leukaemia using the TensorFlow Python programming library (Saeed et al., 2022). As a result, these tools collectively enhance feature extraction and model stability that helped to enhance the blood cancer detection ability of the system considerably.

2.5 Significance and efficiency of multiple image processing techniques in cancer detection

Integration of multiple image processing techniques in cancer detection enables enhanced accuracy, improved diagnosis, and early, automated detection that contribute to overall efficiency. Giri and Saravanakumar (2017), have utilised different image-processing techniques in breast cancer detection including image pre-processing, segmentation, feature extraction and selection, classification technique, and Features for Mammograms. It reflects the multiple images processing technique is integrated into the entire image processing for the medical application for the breast cancer detection process ensuring high efficiency of model performance. In this aspect, the mammogram analysis and its effectiveness and accuracy are effectively optimised through several image processing techniques. In the aspect of detecting blood cancer, such multiple image processing techniques can contribute to the quality enhancement of images, segmenting the meaningful regions, isolation and selection of patterns, and classifying the cancerous cells. On the other hand, Jiang et al. (2023), have evident integration of Convolutional Neural Network (CNN) for image processing where a backpropagation algorithm is utilised for training the model and data augmentation technique utilised for noise reduction, geometric transformation, and others. CNN is a highly efficient and effective technique integrated into image processing that leverages deep learning for automatic and precise feature extraction and delivering suitable image data for medical imaging. Moreover, the data augmentation technique enhances the size and diversity of the training image data thereby contributing to enhanced model performance and robustness of detection and image classification. In the aspect of blood cancer detection, such image processing techniques can extract the molecular and visible data to detect the comprehensive factors from the cancer images. Thus, image processing, segmentation, feature extraction, selection, data augmentation, and CNN are major image segmentation techniques that enable effective cancer detection.

2.6 Benefits of cancer Detection using deep learning models

Utilisation of deep learning models in cancer detection enhances accuracy, efficiency, automation, early detection, and adaptability concerning multiple types of cancers. Deep learning models are effective in learning and detecting complex patterns of cancer images through the effective identification of subtle features (Maurya et al., 2023). In this aspect, the early detection of cancerous cells in image datasets enables more accurate and early diagnosis and treatment plans thereby enhancing healthcare delivery. The integration of ML algorithms encapsulates the

triggering features of cancer present in the image and helps to diagnose the disease in the early stage with a more efficient identification process. In a similar context, Madani, Behzadi and Nabavi (2022), mentioned that the patient survival rate is significantly increased due to the Deep Learning integration in the early cancer detection approach in the medical field. Deep learning utilises multiple layers that support appropriate decision-making regarding cancer detection in complex contexts including blood cancer, breast cancer, brain tumour, and others. However, Madani, Behzadi and Nabavi (2022), have argued that the Deep Learning often can lead to decreased model performance due to irrelevant feature extraction from the images referring to the final label. Thus, it can be stated that adequate implementation of the Deep Learning models delivers numerous benefits including effective decision making, detection accuracy, early identification, and applicability to different cancerous detection.

2.7 Comparison of various deep learning techniques specific to Blood Cancer Prediction

There are different types of blood cancer in the medical scenario, such as myeloma, Leukaemia, and Lymphoma, which affect the blood, bone marrow and lymphatic organs of the body. As per the understanding of Bashiri et al. (2017), the proper prediction of cancers leads to effective and early treatments for improved patient outcomes. In that scenario, Sampathila et al. (2022) have stated that blood cancers are effectively predicted by the different deep-learning models, which automatically identify the complex patterns of different medical data, such as patient data and medical images. In deep learning machine learning model development, the leading models are the Convolutional neural network (CNN), recurrent neural network (RNN), Deep belief network (DBF), Graph Neural Network (GNN), and Hybrid models. Rusnac and Grigore (2022) have recommended that the CNN model can extract the features efficiently from high-quality images and also identify the spatial pattern in the high-quality images. On the other hand, Iman, Arabnia and Rasheed (2023) stated that Transfer learning is also an important feature of the CNN machine learning model. According to Taye (2023), a large amount of data is needed in the CNN and the interpretability of the model is also limited for the CNN model. Further, Salehi et al. (2023) have also stated that the application of the CNN model is generally seen in medical and any kind of image classification. On the other hand, the RNN model is capable of sequential data handling and temporal dependencies. According to Donor Mienye, Swart and Obaido (2024), the RNN model is limited only to the sequential data and there is also huge computational power required to build the model effectively. Transfer learning is also an important part of the Deep learning models and

the pre-trained models are utilised in this type of machine learning predictions. Along with that, the training time is also reduced by transferring data from one domain to another in the application of the Transfer Learning algorithm. As per the consideration of Iman, Arabnia and Rasheed (2023), Transfer learning has been effective, depending on the architectures of the Convolutional neural network, in predicting the possibilities of disease among individuals. The Hybrid models are also an important part of deep learning as they integrate all the deep learning models to create one model. Moreover, in medical diagnosis and disease prediction, CNN model is appropriate as it can handle large datasets with the highest efficiency.

2.8 Challenges faced by cancer detection using deep learning models

2.8.1 Dataset problems and limited high-quality data

Deep learning models excel while trained on large datasets but in the case of blood cancer detection, there is a lack of high quality. According to Jiang et al. (2023), the lack of data has caused over-representation and rendered the computer models to be highly efficient in the training but poorly on unseen data. This inability to accurately predict outcomes on new data is hundred the practical adoption of deep learning models in blood cancer detection that reduced their effectiveness and traceability in improving patient outcomes.

2.8.2 Lack of explainability and model transparency

The Interpretability of deep learning models was another limitation when it came to detecting blood cancer. Ghaderzadeh et al. (2021), stated that blood cancer diagnosis was based on high-level genomics, biomarkers identification and image processing line blood smear images. However, Habehh and Gohel (2021), argued that traditional ML models like decision trees offered greater transparency that allowed clinicians to understand ways certain features contribute to a diagnosis. In contrast, Tran et al. (2021), argued that Deep Learning models, especially those utilising CNNs in classifying blood smear images, were trialled as “black boxes" because the understanding of ways such models arrive at a given decision has been unclear ***[Refer to Appendix 2].*** This lack of transparency made it difficult for healthcare professionals to trust Deep Learning models, especially in critical medical decisions involving cancer treatment.

2.8.3 Adaptation of new data

Generalisation was the biggest drawback of deep learning models for blood cancer detection. As highlighted by Zhang and Hong (2023), a requirement of large, comprehensive and diverse data is needed for Deep Learning models to improve generality across different conditions. This problem

was especially acute in screening for blood cancer, where the ability to generalise across different environments was essential for their broad application in practice.

2.9 Redressal strategies for overcoming challenges in deep learning algorithms

2.9.1 Data augmentation

In order to address the issues of lack of high-quality datasets in blood cancer detection, data augmentation techniques have been utilised to artificially expand the datasets. According to Siebert, Groß and Schroth (2021), enhancement has used rotation, flipping, changing scale and adding noise to test data in order to enhance the features of DL. Furthermore, Arora and Arora (2022), stated that the artificial data generation techniques include production images that signify blood smears using Generative adversarial networks (GANs) have been used to create realistic images. Therefore, these techniques significantly enhanced model robustness and improved performance in detecting blood cancer by addressing a limited dataset.

2.9.2 Integrate explain-ability techniques

The explain-ability techniques have to address the lack of transparency in DL models for blood cancer prediction. Sun et al. (2023), suggested that Shapley Additive Explanations (SHAP) enables one to understand ways in which specific features contributed to the predictions being made. In this way, the interpretability of these models helped clinicians understand certain decisions and thus gain more confidence in such decisions, especially in critical areas of life such as cancer diagnosis.

2.9.3 Continuous training

The ability to maintain and update deep learning models was another important feature of staff training that helped to enhance blood cancer identification. Bhati, Deogade and Kanyal (2023), argued that staff has ensured that the models remain accurate and adaptable by equipping healthcare professionals with skills to handle new data and retain models. For example, NHS AI labs have piloted programs, and integrated AI models for cancer detection using real-time patient data in 2021 (NHS England, 2023). Thus, training increases the efficiency of diagnostics and maintains the technology corresponding to the development of blood cancer detection.

2.10 Theoretical Underpinning

Diffusion of Innovation Theory

The diffusion of innovation theory (DOI) evaluates the expansion and penetration of a new product or technology through a population (Rogers, 2003) (***Refer to figure 4***). The theory states that

different factors including context, innovation characteristics, and adopters significantly influence the adoption of a significant innovation. In this aspect, the DOI theory explains the way of utilising new technologies such as AI-based blood cancer prediction systems adopted by the age population. The impact of the advanced technology is the combined approach of AI and image dataset while detecting blood cancer. The adopters such as diagnostic labs, healthcare sectors, and others can influence the adoption of such technologies based on their awareness and performance.

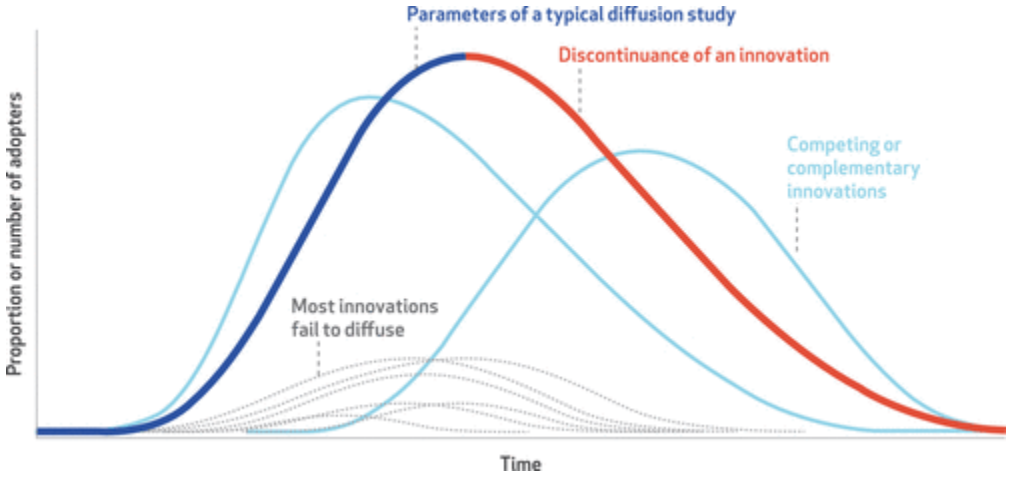

Figure 4:Diffusion of Innovation Theory

(Source: Rogers, 2003)

2.11 Conceptual framework

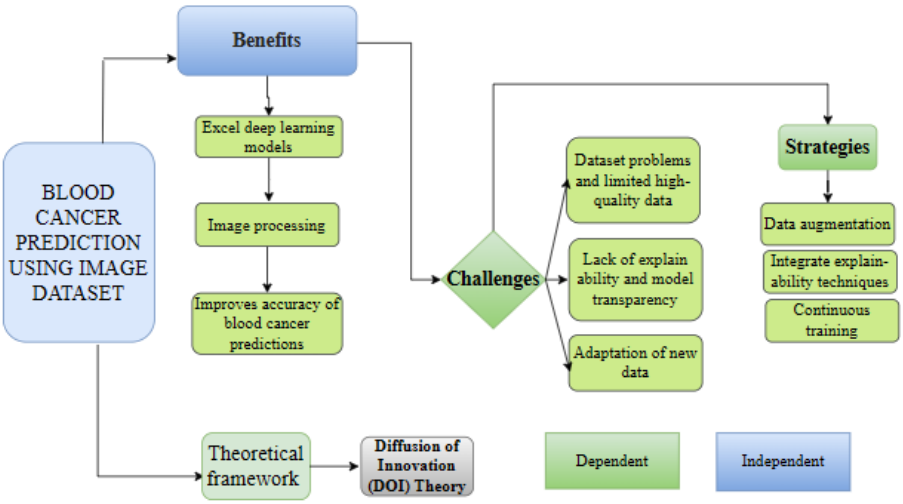

Figure 5:Conceptual framework

2.12 Literature gap

The previous studies have explored the techniques and implementation process of advanced ML models in conventional cancer detection processes that lack proper alignment of clinical and genetic factors. Also, there is not enough focus in real-world setting and making sure findings apple to different subtypes of ALL, namely Pro-B, Early Pre-B and Pre-B. Moreover, the existing paper used augmentation techniques like random cropping, horizontal flipping, vertical flipping and centre cropping. The random cropping and centre cropping will remove critical diagnostic details which will tell the difference between the benign and malignant cells. Not only that, the accuracy and loss of the existing paper shows overfitting model (Ahad *et al.*, 2024).

2.13 Summary

From the study, it has been understood that Deep learning models are effective in the prediction of cancer possibilities from medical images. The deep learning models can effectively predict the possibility of cancer by classifying the images according to the image labels and after training the model automatically predicts the cancer possibility. It has been also understood that machine learning models are more effective than the other conventional methods of cancer detection. In this context, the machine learning models are efficiently developed by the appropriate data pre-processing of the image data by image scaling and normalisation and Exploratory data analysis according to the image labels.

3. Methodology

3.1 Applied Programming Software

The blood cancer prediction in ALL is performed in Jupyter Notebook due to its interactivity, flexibility, and easy-to-use features. The Jupyter Notebook platform enables easy model development, visualisation, and exploratory data analysis in complex contexts, such as cancer cell prediction and visualisation. As mentioned by Negrini, Shabadi and Urban (2023), in machine learning model development Jupyter Notebook is extensively used for precise data preprocessing and developing model-based and interactive datasets. In this aspect, the platform enables the present research to model the raw data easily. It serves as an open-sourced and interactive tool to develop and visualise the results of the machine learning models effectively. Moreover, in this research, Jupyter enables breaking down the complex process of blood cancer prediction into required manageable tasks for easy model implementation and prediction process.

The present research for blood cancer prediction with image datasets such as CT scans and blood smears undergo complex processes of image processing, feature extraction from the image datasets, and integration of ML models for classification. In this context, the cell-based structure of Jupyter allows independent execution of the steps thereby ensuring easy debugging, and refining the steps or codes. Moreover, visualisation is a necessary element in the output file that requires an immediate response with noticeable changes while updating or interpreting the codes in model implementation (Kovač et al., 2024). In this aspect, pattern optimisation and enhancement of model performance are effectively performed in this platform with the feedback of visual representations of blood cells. Therefore, to optimise reliability, flexibility, and reproducibility in sensitive research areas such as blood cancer prediction the Jupyter Notebook is utilised.

3.2 Utilised Programming Language

The study has been used in the Python programming language for blood cancer prediction by analysing an image dataset, which contains the information of acute lymphoblastic leukaemia. As mentioned by Rayhan and Gross (2023), python is a general-purpose programming language that is gradually replacing other languages for research. It has been known for its reputation of being quite basic, uncomplicated as well as versatile to be used in all sorts of applications. Moreover, Raschka, Patterson and Nolet (2020), stated that the relevance of Python is felt based on its simplicity, availability of comprehensive libraries and support for the Python community. Its expensive libraries including Numpy, Pandas, Matplotlib, Seaborn, Tensorflow, and Scikit-learn

were crucial in developing accurate models to have a chance to predict research objectives (Sundaram et al., 2023). These libraries include predefined functions that made model building and testing quite easy in this research. In view of research objectives, the value of Python resulted from its flexibility and applicability in different data types including images, generated data and clinical data. This has been particularly helpful to the objective of the research since Python allowed for easy incorporation of various libraries for data fusion for multimodal data. In addition, the deep learning frameworks implemented in Python allow the capability of making a side-by-side comparison of different algorithms to assess their performance concerning blood cancer prediction. Thus, this programming language has been significant for achieving the research objectives and effectively developing and improving prediction accuracy.

3.3 Utilised Libraries and Framework

Different programming libraries of the Python programming language have been used in machine learning development for the extensive collection of libraries and frameworks, which further help in analysing the image data more efficiently. Basic libraries such as Pandas and Numpy have been utilised for the basic dataset operations. Further, the Pandas have helped construct the dataset from the medical images by defining the image location and class in a table format. Key features of Pandas library are efficiency in reading and writing data from different data formats, such as CSV, and SQL, Handling missing data efficiently, Data slicing and data filtering (Alzoubi, Topcu and Erkaya, 2023). The visualisation libraries from Python have also been used for visualising the image classes in the cancer image dataset. Matplotlib, seaborn and the Plotly library have been used to properly visualise the images. Matplotlib is a foundational library for visualisation, allowing great control over the plots and graphs (Hasija and Chakraborty, 2021). Moreover, Seaborn is created depending on Matplotlib and it is used for creating high-level graphs and interactive visualisations. Common use cases of Matplotlib and Seaborn are the statistical plots, and multi-axes subplots (Vuckovic et al., 2021). In this research, the image data has been analysed by applying different image classifications and regression. In this scenario, Keras and sci-kit-learn are the most used frameworks of Python. The sci-Kit-learn framework has been installed in the machine learning environment by the pip install process. Further, it has helped in efficient data pre-processing and feature extraction. The Keras framework has also been installed following the Pip install process and it has simplified the process of deep learning model building.

3.4 Data Collection Process

The present research is involved in secondary quantitative data collection performed to integrate existing relevant and high-quality datasets for ensuring model and result validation. As mentioned by Jung (2019), the quantitative data enables evidence-based research for effective decision making thereby ensuring valuable insight generation on the research variables. Moreover, as mentioned by Wickham (2019), secondary quantitative data supports clinical research expertise with increased transparency and reproducibility. Secondary datasets contain huge and different sample sizes of different clinical aspects that are necessary for robust machine learning model development. In this aspect, the secondary quantitative data for blood cancer prediction plays an important role as it involves delivering relevant and evidence-based data that provides transparency to the outcomes of prediction models. To ethically integrate the clinical data, Kaggle dataset is utilised to support cost and time-effective analysis and model development. Additionally, a standardisation way of benchmarking a new and robust model for prediction is crucial for enhancing prediction accuracy in medical research which is obtained through secondary quantitative data integration and model tests. However, the primary data and qualitative data include biased findings and a lack of in-depth analysis of results in complex clinical aspects (Pyo et al., 2023; Schneider et al., 2023). Therefore, the primary data and qualitative data are not integrated into the research as it limits the scope and availability of the relevant data for developing predictive models rather secondary quantitative data is utilised for producing reliable, reproducible results. In this regard, this study has included the utilisation of an effective image dataset containing information on acute lymphoblastic leukaemia to predict its occurrence among individuals accurately.

3.5 Data Analysis

3.5.1 Data Cleaning

In this study, python was crucial for data cleaning, and improving accuracy in blood cancer prediction models. Guo et al. (2023), mentioned that Python is used for data cleaning based on a proposed framework addressing duplicate, missing and outlier data. The workflow includes features such as duplicate record removal, data preprocessing, data entry or deletion of missing values, and outlier identification by normalisation and scaling. This has ensured accurate, efficient, and ethical data processing for real-world research. Therefore, this has been stated that Python has

enhanced the quality of the dataset, ensured accurate model training and improved predictive performance in healthcare applications.

3.5.2 Exploratory Data Analysis

Exploratory data analysis has been useful in this study aimed at predicting blood cancer using an image dataset. According to Komorowski et al. (2016), exploratory data analysis enabled the researchers to bring underlying trends and patterns in the data that are crucial for developing robust predictive models. Through the visual interpretation of the dataset utilising Python libraries such as Matplotlib, Seaborn, pandas and Plotly, features were easily detected contributing to improved decision-making in feature selection and modelling. This in turn increased the clarity of the connection between clinical data and image data, thereby increasing the predictive accuracy of the system. Additionally, exploratory data analysis has helped identify anomalies, missing values and outliers that have significantly affected the model performance. Therefore, this data analysis laid the groundwork for successful predictive modelling and directly supported the research questions of improving blood cancer prediction accuracy.

3.5.3 Data Pre-processing

Data preprocessing is the most important step of the machine learning model development, as it prepares the raw data into a suitable format for the machine learning model training. In this process, the machine learning models have been efficiently developed and the model accuracy has been ensured by the proper data preprocessing. Data overfitting has also been prevented by preprocessing the data properly. The data preprocessing step handles the outliers in the data, scales the features and encodes the necessary variables of the data to define the target of the machine learning model (Alshdaifat et al., 2021). In this research, image classification has been performed properly by loading the data by the image data generator of the Keras framework, after that the image has been resized according to the requirement of the machine learning model. Further, the pixel values have been rescaled and the colour spaces have been converted for data visualisation and to identify the classes of images. Data augmentation and normalisation have also been performed in the handling of the imbalances of the data.

3.5.4 Model Selection

The model selection process involved the evaluation of multiple algorithms and the comparison of their performance metrics including precision, accuracy, recall, and others by utilising cross-validation. The viability of the machine learning (ML) models is examined for an effective model

selection for the prediction process that enables higher accuracy in predicting the target variables (Moran et al., 2022). In this aspect, different ML algorithms are implemented to evaluate and compare their performances based on the key metrics for selecting the most effective model for cancer cell prediction in blood. The prime aspect of the model performance involves differentiating the cancerous and non-cancerous cells present in the image dataset which is performed through recognising patterns in the images. Therefore, overall correctness, actual positive instances predicted, and sensitivity of the models for handling complex data are analysed through the key metrics, and based on that the ML model is selected for the blood cancer prediction.

3.5.5 Model Training

The model training process in Machine Learning involves learning patterns and features from a given dataset to optimise the differences between target output and input features for making an accurate prediction on unseen data. The model applies relevant algorithms to the training dataset where adjustments of the parameters such as biases and weights are executed, and differences in the actual target and predictions are managed (Dankers et al., 2018). It reflects the training process significantly including the learning process of the model for outlining accurate predictions. In this context, the ML models are effectively trained to optimise learning patterns, improve accuracy, and develop predictive capabilities regarding the features present in the image dataset of blood cancer prediction in this present research. Therefore, the model training is adequately executed for enabling the ML models to perform cancerous cell prediction based on data-driven decision-making, and prediction with high accuracy and efficiency.

3.5.6 Model Evaluation

The model evaluation step, in the image classification and prediction helps to identify the generalisation ability of the trained model (Lee, 2023). There are different metrics and techniques in the different machine-learning tasks, such as multiclass classification, multilabel classification, and binary classification. The dataset has been split into three datasets train, test and validation set for the proper evaluation of the machine learning model. The training set has been used for training the model. The validation set has been created for tuning the hyperparameters and preventing the overfitting of the model during training. Other performance-measuring metrics, such as cross-validation and confusion matrix have been used for evaluating the machine learning model performance properly. Confusion matrix generally present a detailed breakdown of the performance regarding machine learning by providing the true positive, false positive, true

negative, and false negative (Markoulidakis et al., 2021). The cross-validation accesses the performance of the machine learning model in terms of predicting acute lymphoblastic leukaemia among individuals by analysing the performance of the model depending on the different folds of the dataset.

3.6 Ethics

Ethical considerations were paramount in this study, particularly regarding the use of secondary data analysis. This research has been allowed the ethical standards of the research as provided by the Copyright Act to protect disseminated information. In the Copyright, Designs and Patent Act 1998, the ethical principle of identifying authors and proving acknowledgment to the authors through proper citation of articles as being relevant to conduct research studies (Government of UK, 2024). It was crucial to comply with all legal regulations with university guidelines concerning data usage and privacy to protect data. Social considerations ensured that this research benefited the community while maintaining a professional standard in data handling and reporting practices. Concern has been paid to possible violations of copyright, as the works of other authors have been protected.

4. Data Analysis

4.1 Introduction

The current study has analysed relevant data containing blood reports of patients in order to develop a machine learning model that can effectively predict the possibilities of acute lymphoblastic leukaemia. The Python programming language has been utilized for image recognition and classification. Several python frameworks have been used such as Keras, SKlearn, TensorFlow and others for developing the model.

The data analysis chapter has presented the outcomes of analysis of gathered data to understand the effectiveness of the ML model in predicting acute lymphoblastic leukaemia blood cancer accurately.

4.2 Machine learning development and deployment

4.2.1 Importing libraries and framework

The first step of the machine learning development is to import the libraries and framework for the data visualisation and the machine learning development. The Pandas and Numpy has been imported for creating the data frame and performing the numerical operations in the image database. The TensorFlow and the keras library has been used for the machine learning model development. The Sklearn framework has been used and the metrics has been used from the Sklearn framework. Keras and TensorFlow library has been also used in processing the image data by image scaling techniques. The Keras has been imported in the machine learning environment for importing the prebuilt machine learning models. It has been also seen that the warning has been imported along with the matplotlib and seaborn.

4.2.2 Creating data directory

After importing the required libraries and framework, the data directory has been created by defining the image data location. The image data directory has been created by defining the filename and the directory name and the result output has represented the file path for creating the dataframe.

4.2.3 Data directory according to the image class and image labels

The dataframe has been created by defining the file path of the images and defining the labels of the image. In this dataframe, the labels of the image data are the, Benign, Malignant Pre B,

Malignant Pro B, Malignant Early Type B. The filepath has been defined in the Fseries and the labels of the images has been defined in the Lseries. The blood cell data has been defined in the dataframe by the F series and L series for creating the proper dataframe.

4.2.4 Defining training testing and validation

In total, there are 3234 blood sample images in the dataset. The training set, testing set and the evaluation set shape has been defined and the train_test_split has been imported from the sklearn model selection. In this machine learning model development, the training and testing ration has been split as 70 and 30 respectively by defining the random state as 42. On the other hand, the training set, testing set and the validation set has been created by defining the shape, where the training set has 2587 images, validation set has 647 images and testing set has 971 images.

4.2.5 Data Augmentation and defining images in all sets

Different image augmentation techniques like rotation range, width shift range, height shift range, shear range and horizomtal flip were implemented. The rotation range was set to 20 to rotate the images within 20 degrees randomly. The width shift range and height shift range was set to 0.2 to shift images 20% horizontally of the width and vertically of the height respectively. The shear range was set to 0.15 to set shear angle between -0.15 and 0.15 whereas the horizontal flip was set to true to flip the image horizontally. The augmented image data generator was applied to the training dataset and a image data generator is applied to the validation and testing dataset without the augmentation.

Filepath and labels were set to the x and y columns respectively. The size of the image was set to 244 x 244 pixels. RGB color mode is maintained since even small changes in pigmentation is crucial to determine the differences between different subtypes of ALL. The class mode was set to categorical itself, the maximum size of a batch was set to 25 and the dataset was not shuffled. These specifications was commonly applied to training, validation and testing set and has been created properly by defining the different number of images from 4 classes.

After defining the training, testing and the validation set, the cancer images from different set has been visualised for ensuring the consistency of the image quality. The visualisation of the image

data has been made by defining the show image function. Along with that, the If else statement has been used in the program for defining the length of the image data, which is same as that of batch size. Further, the matplotlib library has been defined in the machine learning environment for creating the subplot for the images.

4.2.6 Inception-V3 Model Training

The first deep learning model is the Inception V3 model by defining the input shape as (244,244,3). The base layer of the pre-trained model was freezed for the model to learn parameters from the custom layer. In the base layers of the deep learning model, the inception layers have been defined and the base layers are fixed. Therefore, the custom classification layers have been trained by defining the Dense as 1024, batch normalization and the dropout as 0.5 to drop 50% of the parameters. The second layer have been trained dy defining the dense as 512, batch normalization and the dropout as 0.5. Relu activation function was used to define both the layers to introduce non-linearity. The final layer have ben trained by defining dense as 4 for the four classes with softmax activation function to predict the probabilities. The Inception V3 model has been compiled by defining the Optimiser with adam optimizer with a learning rate of 0.0001, loss with categorical crossentropy since there exist more than two classes and the metrics with accuracy. In the deep learning model, the 5 epoch has been defined for the validation of the data, where the verbose has been defined as 1 to show the history of each epoch. The training accuracy of the machine learning model was 91% and a validation accuracy of 95% has been achieved at the end of 5 epoch.

4.2.7 ResNet50 Model Training

The ResNet50 model has been trained as the second model for training. The input shape of the ResNet model has been given (244, 244, 3). The base layer of the pre-trained model was freezed. The custom classification layers have been trained by defining the dense, batch normalization and dropout. The formation of the custom classification layers is same as that of the previous model. This model has also been compiled by defining the Optimiser with adam optimizer with a learning rate of 0.0001, loss with categorical crossentropy and the metrics with accuracy. The training accuracy of the machine learning model was 78% and a validation accuracy of 43% has been achieved at the end of 5 epoch.

4.2.8 MobileNet-V2 Model Training

The MobileNet-V2 model has been trained as the third model for training. The input shape of the model has been given (244, 244, 3). The base layer of the pre-trained model was freezed. The custom classification layers have been trained by defining the dense, batch normalization and dropout. The formation of the custom classification layers is same as that of the previous model. This model has also been compiled by defining the Optimiser with adam optimizer wit a learning rate of 0.0001, loss with categorical crossentropy and the metrics with accuracy. The training accuracy of the machine learning model was 94% and a validation accuracy of 97% has been achieved at the end of 5 epoch.

4.2.9 Custom CNN Model Training

The CNN model has been trained as the fourth model for training, where the deep neural network has been designed by defining the three convolutional layers with filters as 64, 128 and 256. The batch noirmalization has been applied to normalize the output from the previous layer to increase the training speed. The pool size in the Max pooling layer has been defined as (2,2) followed by a dropout of 0.5 to to drop 50% of the neurons in the layer to prevent overfitting. A dense layer of 512 has been defined followed by a final dense layer of 4 neurons with relu and softmax activation functions respectively. In the model compilation step, the categorical crossentropy has been defined for the loss of the machine learning model. In the custom history configuration of the model, 5 epochs has been defined for generating accuracy from the machine learning model. The training accuracy of the machine learning model was 92% and a validation accuracy of 46% has been achieved at the end of 5 epoch.

4.3 Evaluation of the MobileNet-V2 Model

The machine learning model has been evaluated by the evaluation matrix, where the accuracy score of 4 models has been presented. The testing accuracy of the InceptionV3, ResNet50, MobileNetV2 and custom CNN models was 95%, 45%, 97% and 47% respectively. According to the result output, the MobileNetV2 model has the highest accuracy score. *(Refer to Figure 6)*

```
Inception-V3 Evaluation:
39/39 ━━━━━━━━━━━━━━━━━━━━ 240s 3s/step - accuracy: 0.9536 - loss: 0.1487
ResNet50 Evaluation:
39/39 ━━━━━━━━━━━━━━━━━━━━ 168s 2s/step - accuracy: 0.4553 - loss: 2.2479
MobileNet Evaluation:
39/39 ━━━━━━━━━━━━━━━━━━━━ 93s 2s/step - accuracy: 0.9708 - loss: 0.0699
Custom CNN Model Evaluation:
39/39 ━━━━━━━━━━━━━━━━━━━━ 126s 2s/step - accuracy: 0.4722 - loss: 16.6895
```

Figure 6: Evaluation of the machine learning models

Since MobileNetV2 is the best fitting model with a testing accuracy of 97 percent, classification report of the MobileNetV2 has been defined ***(Refer to Figure 7).*** The model achieved an exceptional precision, recall and f1-score of 0.98. This proves that the model identified 98% of the true positive and actual positive values.

	precision	recall	f1-score	support
Benign	0.98	0.92	0.95	173
Malignant_Pre-B	1.00	0.99	1.00	281
Malignant_Pro-B	1.00	0.99	1.00	226
Malignant_early Pre-B	0.94	0.99	0.97	291
accuracy			0.98	971
macro avg	0.98	0.97	0.98	971
weighted avg	0.98	0.98	0.98	971

Figure 7: Classification report of MobileNetV2

The model accuracy and model loss graph between train and validation set have been defined to understand the model performance during training. From the figure, it is understood that both the training and validation curve is having a steady increase and decrease for the accuracy and loss curve respectively and the difference in accuracy is less than 3 percent. This proves that the model is not overfitting. ***(Refer to Figure 7)***

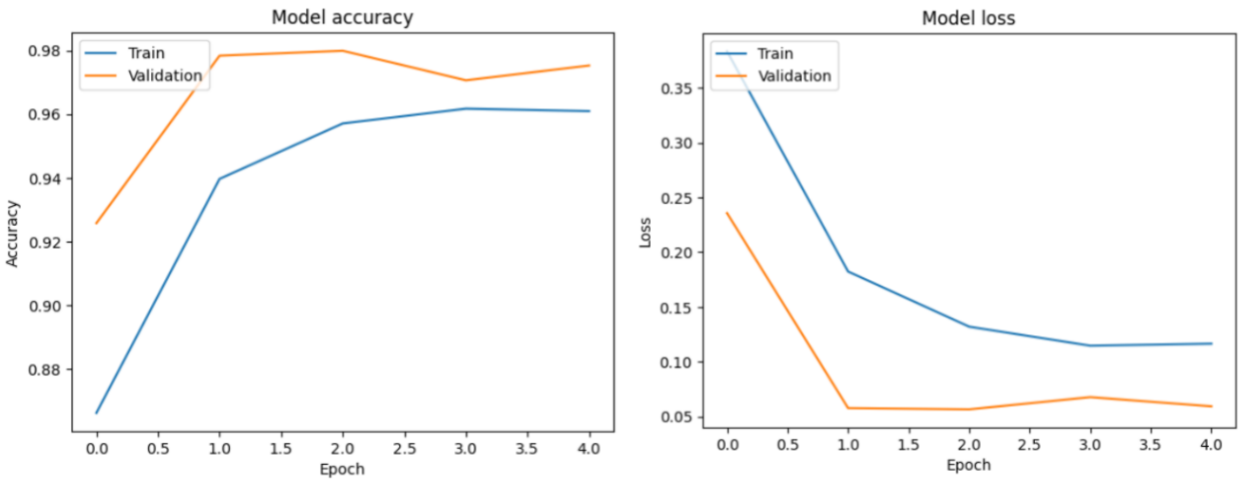

Figure 8: Accuracy/Loss curve of MobileNetV2

In the next step the MobileNet V2 model has been implemented in the image prediction. In order to predict the blood cell image, the MobileNetV2 model has been implemented in the GUI where the model learn the classes from the image and predict any blood sample image by classifying into the predefined classes.

4.4 Graphical User Interface

The graphical user interface has been created for the blood cancer prediction by using the Tkinter library from python. In the first step, the classification class has been used for the MobileNetV2 model by defining the file path for importing the image. The MobileNetV2 model in this task, predict the image by analysing the previous image array and defining the thresholds for the detection of invalid detection. The last function of the image prediction is the image uploading and classifying array. It has been also defined that the interface is going to show the image and the predicted class below the Image. The last step of the prediction interface development is to create the Graphical user interface by combining the image uploading button, image label and the result output label.

The prediction output has been seen in the Figure 8, which has been created by python Tkinter library. The result output of the prediction interface has shown that there is an option for uploading the image and the representation of present image along with the predicted class of the image. The upload image button provides the opportunity to upload the image from the local computer. After that, the MobileNetV2 model classify the images applying the algorithm and predict the class

correctly for the particular image. Similar predictions were also performed to the other subtypes of ALL and all the predictions were correct. ***(Refer to Figure 8)***

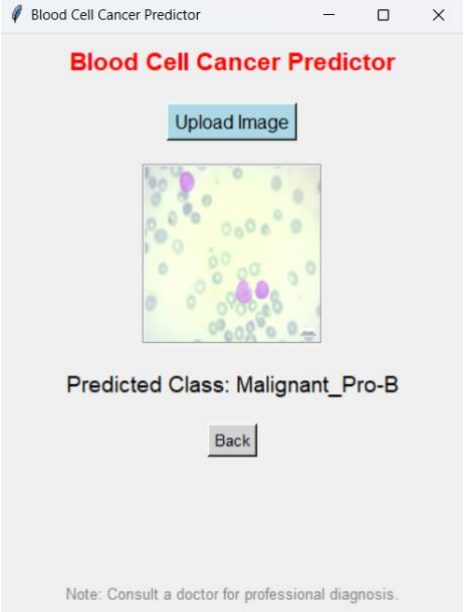

Figure 9: Prediction output of Malignant Pro-B

4.5 Invalid Data Prediction

If any image which is not related to a blood cell sample was given to the user interface to check the prediction, the interface will not predict as one of the predicted class in the dataset. Instead, the interface will show as “Invalid Image”. ***(Refer to Figure 9)***

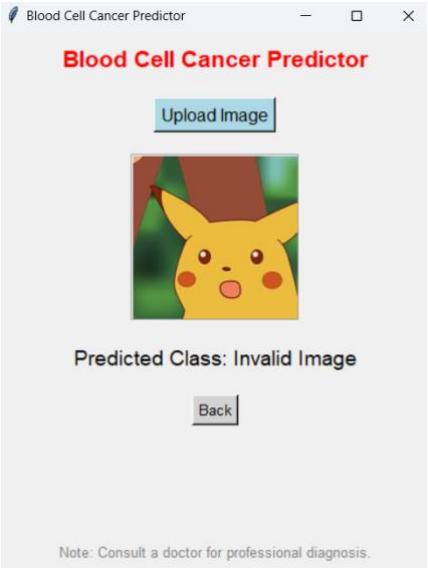

Figure 10: Prediction output of Invalid Image

4.5 Summary

The deep learning model has been developed in this image classification task for the proper prediction of the blood cancer. The Blood cancer image dataset is divided into 4 class, such as Malignant early Pre-B, Malignant Pro-B, Malignant Pre-B and Benign of acute lymphoblastic leukaemia. In order to classify and predict the image class, 4 machine learning model has been developed such as CNN, ResNet50, MobileNetV2 and the InceptionV3. Among all the machine learning models the MobileNetV2 has performed best with the 97% accuracy, The prediction GUI has been built depending on the MobileNetV2 model and the interface has successfully predicted the image class of acute lymphoblastic leukaemia.

5. Findings and Discussions

5.1 Introduction

The current study has been executed to investigate appropriate solutions for predicting the possibilities of acute lymphoblastic leukaemia by analysing several blood cell images using machine learning models. Python programming language has been used due to the easy application of it in machine learning. Several built-in libraries and frameworks of Python have been utilised as per requirement followed by data processing and data splitting to achieve the highest accuracy in the machine learning model.

5.2 Findings

The machine learning model has been developed for predicting cancer by using different libraries and frameworks from the Python programming language. In the study, four deep learning algorithms, InceptionV3 model, ResNet50, MobileNetV2 and a Custom Deep CNN have been evaluated for investigating precise solutions for acute lymphoblastic leukaemia prediction. Based on the evaluation, it has been found that the MobileNetV2 algorithm reveals more accurate outcomes with fewer errors than other deep learning models in terms of InceptionV3 model, ResNet50, and Custom Deep CNN. Different in-built libraries, such as Pandas and Numpy, are quite effective in creating data frames and performing numerical operations as they contain numerous functions for complex mathematics. In this context, image data collection of acute lymphoblastic leukaemia is properly visualised using the available Python numerical functions. In the present scenario, early detection of acute lymphoblastic leukaemia has become inevitable due to the increased count of acute lymphoblastic leukaemia cases to many folds. The early detection of acute lymphoblastic leukaemia by using automation especially for Leukemia utilised machine learning through a series of steps such as designing, placing, collecting, analysing, constructing, training and evaluation of data. The incorporation ML algorithm summarises the triggering properties of acute lymphoblastic leukaemia present in the image for early diagnosis of acute lymphoblastic leukaemia in a more efficient manner. As a consequence, the survival rate of people is also significantly increased resulting in lowered overall mortality rate. The ML model has been prepared by preprocessing the data through image generator and scaling the class mode according to the class label. It has also been found that the data set has been transformed to a data frame by defining the labels and location of the image in a local computer before using in the machine learning environment.

5.3 Discussion

5.3.1 Machine Learning Model for Blood Cancer Prediction

The development of a machine learning model enhances the accuracy of the early detection of acute lymphoblastic leukaemia. Early diagnosis of blood cancer by using Artificial Neural Networks (ANN) is a fast and cost-effective technique of early diagnosis of cancer that facilitates timely intervention by doctors (A et al., 2023). Further, Convolution Neural Network (CNN) is quite effective in image processing that possesses the property of more precise feature extraction resulting in more accurate medical imaging (Zhao et al., 2024). On the other hand, the RNN model can handle sequential data only and needs a huge computational power to develop the model properly (Mienye, Swart and Obaido, 2024). However, as per the current study, it has been found that the MobileNetV2 model development is most effective for diagnosing acute lymphoblastic leukaemia efficiently and quickly. Also, the prediction outcome has been influenced by the implementation of the ML model. The key significance of the ML model is enabling the patients to take prior action by identifying the chances of acute lymphoblastic leukaemia in the early stage. In addition to that, it also helps find the mitigation steps of cancers. In the current work, the machine learning model has been successfully implemented by properly utilising libraries and framework from Python. The MobileNetV2 model has been used in the current work to obtain the most accurate result. The model has been effectively developed by using the Sklearn, Keras and TensorFlow framework of Python. Sklearn is mainly built on Numpy, Scipy and Matplotlib which is an effective tool for predictive data analysis. Preprocessing of image data has been performed by the data augmentation and image generator library available in Sklearn and scaling of the acute lymphoblastic leukaemia image data is done properly. *Thus, different Python programming libraries and proper data handling processes have provided effective assistance to the development of ML models for predicting acute lymphoblastic leukaemia among individuals.*

5.3.2 Multiple Image Processing technique for boosting accuracy in Blood Cancer Prediction

Several available image processing techniques for boosting the accuracy of blood cancer prediction are ANN, CNN, MobileNet, Inception V3 and many more. As stated by Andon and Covatariu (2022), image processing using ANN consists of multiple sequential steps such as image processing, data reduction, segmentation, recognition, contrast adjustment, noise reduction, feature extraction and others that togetherly generate an accurate result. On the other hand, as mentioned by Grassa, Gallo and Landro (2021), CNN possesses a hierarchical nature that is

capable of identifying complex features from input images. Therefore, it is supported broadly that CNN provides the most precise medical image processing enhancing the accuracy of early prediction of cancer. As described by Saeed et al. (2022), the EfficientNetV2S model can accurately predict acute lymphoblastic leukaemia by analysing image datasets. However, based on the findings of the current study, it has been noticed that the effectiveness of MobileNetV2 model for accurate early diagnosis is reliable and possesses less error. MobileNetV2 is an efficient model that can recognize images accurately resulting in accurate analysis. One of the noticeable advantages of using MobileNetV2 model for pre-detection of cancer in terms of acute lymphoblastic leukaemia (ALL) is that it can be easily accessed through mobile minimising the need for large setup. *Therefore, MobileNetV2 has been considered the best model to generate accurate outcomes as compared to other available models in diagnosis of cancer (acute lymphoblastic leukaemia) in the early stage.*

5.3.3 Incorporation of Combined Genetic and Clinical Data in Image Data Set For Improving Accuracy of Cancer Prediction

The integration of combined genetic and clinical data influences the accuracy of early prediction of cancer as well as predicting the recurrence of cancer in people who have recovered previously. According to Wijewardhane, Dressler and Ciccarelli (2021), genetic changes that lead to the transformation of normal cells into cancer cells are usually inherited by generation after generation in case of inherited cancer. Therefore, feeding genetic data to the model plays a significant role in enhancing the accuracy of acute lymphoblastic leukaemia prediction in the early stage. On the other hand, clinical data involving cellular information of blood is very important in identifying the chances of cancer in people who have experienced a recovery from the disease in a previous stage of life (Das et al., 2023). Proper clinical data is important for detecting the possibilities of the disease in the early stages and the initiation of the treatment in early stage prevent the severeness of the disease. Any minor changes in those factors can be a significant sign of acute lymphoblastic leukaemia in later phases of life. *Therefore, the ML model provided with both genetic as well as clinical data boosts the accuracy of detecting chances of acute lymphoblastic leukaemia in people in the early stage, primarily preventing the mortality rate.*

5.3.4 Assessment of Different Deep Learning Algorithms in Terms of Accuracy in Acute Lymphoblastic Leukaemia Prediction

The deep learning model has helped in evaluating the data set accurately by providing better data visualisation in the early diagnosis process. The study has assessed different deep learning algorithms such as InceptionV3 model, ResNet50 algorithm, MobileNetV2 algorithm and Custom Deep CNN. Evaluation of the outcomes of the mentioned ML models have revealed that the MobileNetV2 algorithm is most effective in diagnosing acute lymphoblastic leukaemia in the early stage. The model is found to have better accuracy as compared to other ML models with negligible error whereas the outcome generated by the ResNet50 model is found to be least accurate with noticeable loss due to error. MobileNetV2 has the feature of low latency, easy adaption into any application, low power and others that results in a balance between accuracy of the model and computational efficiency (Khasoggi, Ermatita and Samsuryadi, 2019). In addition to that, MobileNetV2 also possesses a higher level of accuracy in classification and uses fewer parameters. Consequently, the algorithm can be easily adapted and accessed minimising the level of complexity. *Hence, the use of deep learning algorithms has helped in developing a MobileNet model for efficient image processing and accurate prediction of blood cancer in the early stage.*

5.4 Summary

The chapter has attained research objectives by supporting the contents from articles built through previous research regarding the image processing by machine learning for early detection of acute lymphoblastic leukaemia. The use of deep learning algorithms, MobileNetV2 model and specific frameworks from Python has resulted in developing the desired machine learning model that is capable of accurately identifying the chances of acute lymphoblastic leukaemia in people. It can potentially reduce the increased mortality rate due to late diagnosis and late treatment.

6. Conclusion and Recommendation

6.1 Conclusion

The current research has helped in developing as well as implementing a machine learning model for achieving precision in early detection of acute lymphoblastic leukaemia. The research has evaluated the result of a number of ML models such as the InceptionV3 model, Custom Deep CNN, ResNet50 and MobileNetV2 to reach a reliable solution that can reduce mortality rate due to cancer. Among a number of deep learning algorithms, MobileNetV2 is found to be best in medical image processing in terms of accuracy and least error. Custom Deep CNN, InceptionV3 model and ResNet50 are also applicable in identifying minute details of the medical images as each of them possess numerous unique features that ensures perfection in processing blood reports. CNN possesses the property of feature extraction that boosts the image processing step to enhance the accuracy. It also involves hierarchical nature that enables extraction of complex features from a medical image. Also, the preprocessing of data is usually done by data augmentation for flipping and rotating the images and image generator for scaling the class mode and the RGB scale of the image. Therefore, it has been largely accepted as an efficient way of image processing by many people.

RNN requires large computation power to deal with only sequential information. However, MobileNetV2 uses very few parameters such as width multiplier that helps narrowing a network and possesses a higher level of accuracy in terms of acute lymphoblastic leukaemia. In addition to that, the model can be easily accessed through mobile resulting in less complexity in adaption of the model by the user. Based on the overall study, it can also be inferred that incorporation of the genetic as well clinical data plays a significant role in the early detection of cancer process. Along with that, it also helps to anticipate any chance of recurrence of acute lymphoblastic leukaemia for the patients who have recovered from the disease in the early phase of life.

The model has been developed using Python programming language and Jupyter notebook environment. The simple syntax, predefined libraries and frameworks helped implement the model with less complexity. Also, the language has been used several times for working on machine learning previously. As a consequence, knowledge-based guidance was easily available for developing the model. The model has utilised several built-in libraries and frameworks of the Python programming language. For instance, Pandas and Numpy helped in creating data frames and executing complex numerical operations as they contain a collection of predefined functions

that are capable of solving complex mathematical problems. Frameworks like Sklearn, Keras and TensorFlow have been used for improving the visibility of graphical displays in the process of image processing. The creation and execution of the code has been performed on Jupyter notebook due to the availability of the feature of documentation and sharing in this platform. The platform acts as an open source, interactive tool to create and visualise the outcomes of machine learning models properly. In the current study, the platform helped in subdividing the entire complex process of early prediction of acute lymphoblastic leukaemia into required manageable tasks for easy implementation and execution of the process. Also, the feature of independent step by step execution of the code is ensured by the cell-based structure of the platform. This feature helped in easy debugging and refining the code to meet the desired objectives accurately and quickly. The overall outcome of the study has revealed that the MobileNetV2 model is very much effective in diagnosing cancer accurately in the early stage as compared to other models. As a result, the better visualisation of image data has contributed to early detection which in turn has resulted in the right time treatment by the medical practitioners enhancing the overall healthcare system.

6.2 Limitation

The current research has utilised only secondary quantitative information to create the model that accurately predicts cancer in the early stage. The research has worked with only four models namely Inception V3, ResNet, MobileNet and Custom Deep CNN. It has not utilised other available ML models such as ANN, RNN and others that could enhance the evaluation process in a better way. Moreover, the study has explored only a restricted section of the context that is medical image processing for predicting acute lymphoblastic leukaemia only in the early stage. The study could be extended to the evaluation of the effectiveness of other medical equipment also that are significant for early detection of different medical complications along with cancer. Using more advanced models of deep learning such as Transformers, Vision Transformers can help to achieve a higher accuracy to identifying subtle abnormalities in blood smear images.

6.3 Recommendation

The research may use mixed methods containing both quantitative as well as qualitative information to reach a robust conclusion regarding the effectiveness of ML model in acute lymphoblastic leukaemia prediction. Moreover, considering the real time instances may strengthen the justification of the conclusion, boosting the validity of the research. The collection of human-centric data directly from primary sources may enhance the accuracy of the research outcome

resulting in precise understanding about effectiveness of ML model implementation for predicting acute lymphoblastic leukaemia.

6.4 Future scope for the research

Development of the models for acute lymphoblastic leukaemia prediction by utilizing the image dataset is one of the rapidly evolving fields with an immense potential of revolutionizing the process of treatment and diagnosis. The future research may focus on addressing the potential risk factors and mitigation strategies for developing and implementing ML models in predicting acute lymphoblastic leukaemia in the early stage along with medical image processing. The future research may also help conceptualise the knowledge regarding the differences between normal blood cells and cancerous cells to enhance the accuracy of acute lymphoblastic leukaemia diagnosis in a significant way. Further, the application of machine learning in diagnosis as well as treatment process can be a vast area to be researched. It can facilitate in evaluating both the detection as well as treatment process resulting in better healthcare.

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Appendices

Appendix 1: Used of ML model in cancer prediction

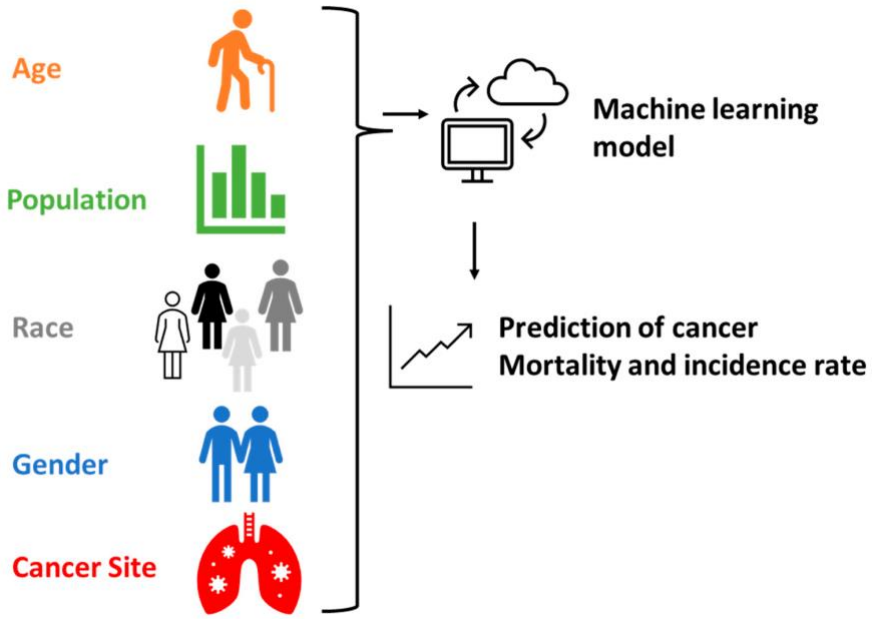

Figure 11: Used of ML model in cancer prediction

(Source: Ding, 2024)

Appendix 2: Deep learning for blood cancer prediction

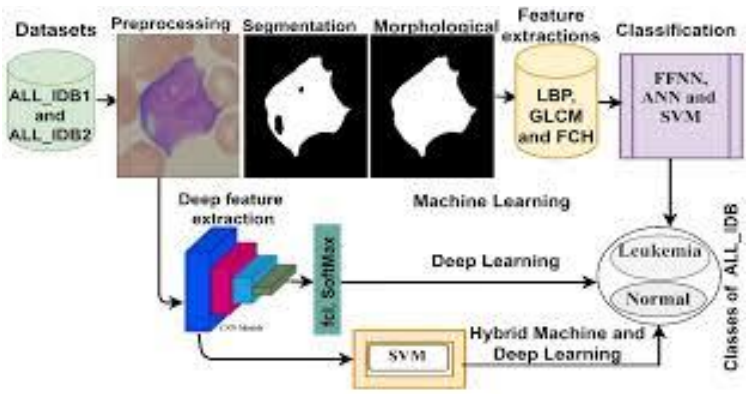

Figure 12: Deep learning for blood cancer prediction

(Source: Abunadi and Senan, 2022)